PERMUTATION, COMBINATION & PROBABILITY

1. There are 12 students in a class. If a group of 5 students has to be selected from the class in such a way that one particular student is always included, in how many different ways can the selection be made?

(1) 210 (2) 220 (3) 320 (4) 330 (5) 420

2. If the letters of the word PREVIOUS be arranged at random, what is the probability that all the vowels come together?

(1) $\frac{1}{8}$ (2) $\frac{7}{8}$ (3) $\frac{1}{14}$ (4) $\frac{7}{14}$ (5) $\frac{1}{16}$

3. Letters of the word "SERIES" are arranged in such a way that all the vowels always come together. What is the total number of ways of making such an arrangement?

(1) 720 (2) 180 (3) 144 (4) 72 (5) 36

4. From a pack of 52 cards, two cards are drawn at random. What is the probability of both of them being queens?

(1) $\frac{200}{221}$ (2) $\frac{20}{221}$ (3) $\frac{201}{221}$ (4) $\frac{1}{221}$ (5) $\frac{32}{221}$

5. The ratio of the number of girls to the number of boys is the 5 : 2 in a class of 28 students. A group of three students is to to be selected at random amongst them. What is the probability that the selected group of students contain one boy and two girls?

(1) $\frac{14}{71}$ (2) $\frac{14}{117}$ (3) $\frac{20}{71}$ (4) $\frac{20}{117}$ (5) None of these

6. 18 persons are sitting around a circular table. In how many ways can they be seated if six particular persons are to always sit together?

(1) 18! × 6! (2) 17! × 5! (3) 13! × 6! (4) 12! × 6! (5) 12! × 5!

7. In how many ways can the letters of the word "PRODUCT" be arranged so that all the vowels never come together?

(1) 5040 (2) 720 (3) 4320 (4) 1440 (5) 3600

8. There are seven boys and six girls. They sit in a row randomly. What is the probability that all the girls do not sit together?

(1) $\frac{1}{429}$ (2) $\frac{428}{429}$ (3) $\frac{2}{429}$ (4) $\frac{427}{429}$ (5) $\frac{1}{1716}$

9. There are seven boys and four girls in a row. In how many ways can they be seated in the row so that all the girls do not sit together?

(1) 7! × 4! (2) 11! - 7! × 4! (3) 11! - 7! (4) 11! - 4! (5) 11! - 8! × 4!

10. A bag contains four red, six black and and seven green balls. Three balls are drawn randomly. What is the probability that the balls drawn contain exactly two red balls?

(1) $\frac{39}{340}$ (2) $\frac{301}{340}$ (3) $\frac{55}{408}$ (4) $\frac{353}{408}$ (5) $\frac{54}{91}$

11. In how many ways can 6 boys and 5 girls be seated in a row so that they sit alternately?

(1) 43200 (2) 86400 (3) 21600 (4) 840 (5) 720

12. When two dice are thrown, what is the probability that the difference of the numbers shown by them will be 3?

(1) $\frac{2}{9}$ (2) $\frac{1}{36}$ (3) $\frac{1}{6}$ (4) $\frac{1}{18}$ (5) None of these

13. In how many different ways can the letters of the word "PRODUCTION' be arranged so that the vowels always come together?

(1) 15120 (2) 30240 (3) 60480 (4) 120960 (5) $\frac{11!}{2!}$

14. From a group of 8 men and 5 women, in how many different ways can six of them be included so that at least one woman is chosen?

(1) 1715 (2) 1688 (3) 896 (4) 996 (5) 1714

15. In how many ways can you choose a group of four persons out of three girls and seven boys such that the group has only one girl?

(1) 105 (2) 210 (3) 120 (4) 224 (5) 240

16. A card is drawn at random from a pack of 52 cards. What is the probability that the card drawn is either a Diamond or a Jack?

(1) $\frac{1}{13}$ (2) $\frac{2}{13}$ (3) $\frac{3}{13}$ (4) $\frac{4}{13}$ (5) $\frac{5}{13}$

17. In a party, every person shakes hands with every other person. If there occur a total of 351 handshakes in the party, how many persons were present at the party?

(1) 25 (2) 26 (3) 27 (4) 28 (5) 29

18. A bag contains 10 white, 8 black and 6 blue balls. If two balls are drawn at random, find the probability of both the balls being black?

(1) $\frac{1}{3}$ (2) $\frac{1}{12}$ (3) $\frac{7}{69}$ (4) $\frac{9}{69}$ (5) None of these

19. In how many different ways can the letters of the word "SCHOOL" be arranged?

(1) 720 (2) 360 (3) 180 (4) 120 (5) 60

20. A card is drawn from a pack of 52 cards. What is the probability that it is either a Queen or a Jack?

(1) $\frac{1}{13}$ (2) $\frac{1}{26}$ (3) $\frac{2}{13}$ (4) $\frac{3}{13}$ (5) None of these

21. From a pack of 52 cards, two cards are drawn at random. What is the probability that one is a Queen and the other is an Ace?

(1) $\frac{1}{221}$ (2) $\frac{32}{221}$ (3) $\frac{20}{663}$ (4) $\frac{8}{663}$ (5) None of these

22. If a total of nine students appear in an examination, in how many ways can the result be announced?

(1) 1024 (2) 512 (3) 256 (4) 2048 (5) 128

23. How many numbers between 300 and 800 can be made by using digits 2,4,5,6 and 0?

(1) 36 (2) 72 (3) 144 (4) 108 (5) None of these

24. If two dice are thrown simultaneously, what is the probability that the sum of the numbers appeared is less than seven?

(1) $\frac{7}{12}$ (2) $\frac{5}{12}$ (3) $\frac{7}{36}$ (4) $\frac{5}{18}$ (5) $\frac{5}{36}$

25. From a group of 8 boys and 6 girls, in how many ways can a group of 5 be chosen so as to include at least one girl?

(1) 240 (2) 1260 (3) 1720 (4) 1840 (5) 1946

26. A bag contains 5 red and 8 black balls. A ball is drawn out of it and replaced in the bag. Then a ball is drawn out again. What is the probability that both balls are red?

(1) $\frac{5}{13}$ (2) $\frac{8}{13}$ (3) $\frac{25}{169}$ (4) $\frac{64}{169}$ (5) None of these

27. How many different arrangements can be made from the letters of the word "WEDNESDAY" such that all vowels come together?

(1) 7560 (2) 15120 (3) 2520 (4) 5040 (5) None of these

28. Seven persons are sitting around a round table. What is the probability that three particular persons are sitting together?

(1) $\frac{1}{7}$ (2) $\frac{2}{7}$ (3) $\frac{1}{5}$ (4) $\frac{2}{5}$ (5) $\frac{1}{6}$

29. In a party, every member shakes hands with every other member. If the total number of handshakes is 561, how many members are there in the party?

(1) 36 (2) 35 (3) 34 (4) 33 (5) 32

30. In a throw of two dice, what is the probablity that the sum of numbers appeared is greater than or equal to eight?

(1) $\frac{5}{36}$ (2) $\frac{5}{18}$ (3) $\frac{5}{12}$ (4) $\frac{1}{6}$ (5) None of these

31. A bag contains 4 Red, 8 Black and 12 White balls. Three balls are drawn randomly. What is the probability that the balls drawn are of different colours?

(1) $\frac{12}{253}$ (2) $\frac{16}{253}$ (3) $\frac{24}{253}$ (4) $\frac{48}{253}$ (5) $\frac{205}{253}$

32. How many different words can be formed with the letters of the word 'HINDUSTAN' so that all the vowels come together?

(1) 30240 (2) 15120 (3) 7560 (4) 362880 (5) 181440

33. In a group of 12 persons, 5 are females. In how many different ways can they sit in a row so that no two females sit together?

(1) 3386.88×10^3 2) 3386.88×10^2 3) 3386.88×10^4 4) 3386.88 5) None of these

34. 12 persons are to be seated around a round table. What is the probability that 5 particular persons sit together?

(1) $\frac{2}{33}$ (2) $\frac{1}{66}$ (3) $\frac{4}{65}$ (4) $\frac{3}{65}$ (5) $\frac{26}{65}$

35. In how many different ways can the letters of the word THERAPY' be arranged so that the vowels never come together ?

(1) 720 (2) 1440 (3) 5040 (4) 3600 (5) 4800

36. A bag contains 13 white and 7 black balls. Two balls are drawn at random. What is the probability that they are of the same colour ?

(1) $\frac{41}{190}$ (2) $\frac{21}{190}$ (3) $\frac{59}{190}$ (4) $\frac{99}{190}$ (5) $\frac{77}{190}$

37. How many numbers between 300 and 1000 can be made with the digits 1, 2, 3, 4, 5, 6, 7 and 0?

(1) 280 (2) 210 (3) 5040 (4) 300 (5) None of these

38. 15 persons are seated around a table. What is the probability that four particular persons always sit together?

(1) $\frac{1}{546}$ (2) $\frac{1}{91}$ (3) $\frac{4}{455}$ (4) $\frac{1}{2184}$ (5) None of these

39. What numbers between 500 and 1900 can be made with the digits 3, 4, 5, 6, 7 and 0 (Repetition of digits is not allowed) ?

(1) 60 (2) 90 (3) 120 (4) 150 (5) None of these

40. From a pack of 52 cards, 3 cards are drawn. What is the probability that it has all three kings?

(1) $\frac{1}{52}$ (2) $\frac{16}{5525}$ (3) $\frac{1}{5525}$ (4) $\frac{48}{5525}$ (5) None of these

41. In how many different ways can the letters of the word PRODUCTION be arranged in such a way that all the vowels always come together?

(1) 362880 (2) 1814400 (3) 10080 (4) 60480 (5) 120960

42. Six boys and seven girls are made to sit in a row. What is the probability that all the boys do not sit together?

(1) $\frac{1}{429}$ (2) $\frac{2}{429}$ (3) $\frac{428}{429}$ (4) $\frac{427}{429}$ (5) None of these

43. In how many ways can six different things be divided equally among three persons?

(1) 3375 (2) 1125 (3) 375 (4) 90 (5) 22

44. A box contains 8 Red, 4 White and 3 Blackballs. Three balls are drawn randomly from the box. What is the probability that the three balls drawn randomly are of different colours?

(1) $\frac{3}{91}$ (2) $\frac{3}{455}$ (3) $\frac{1}{5}$ (4) $\frac{91}{455}$ (5) $\frac{96}{455}$

45. From a group of five males and six females, in how many ways can four be chosen to include exactly one female?

(1) 210 (2) 180 (3) 120 (4) 80 (5) 60

46. A bag contains 6 red, 7 blue and 8 green balls. Three balls are drawn randomly. What is the probability that the balls drawn contain exactly two blue balls?

(1) $\frac{147}{665}$ (2) $\frac{518}{665}$ (3) $\frac{54}{455}$ (4) $\frac{44}{455}$ (5) $\frac{401}{455}$

47. In how many different ways can the letters of the word 'REPLACE' be arranged?

(1) 2630 (2) 5040 (3) 1680 (4) 2580 (5) None of these

48. How many words can be formed from the letters of the word REGULAR?

(1) 1260 (2) 5040 (3) 2520 (4) 720 (5) 2880

49. A bag contains 3 black and 6 white balls. Two draws of one ball each are made without replacement. What is the probability that one ball is black and the other is white?

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$ (3) $\frac{1}{6}$ (4) $\frac{2}{9}$ (5) $\frac{7}{9}$

50. In how many ways can 6 Americans, 5 Indians and 3 Russians be seated in a row so that all the persons of the same nationality sit together?

(1) 1555200 (2) 777600 (3) 1036800 (4) 3110400 (5) 518400

51. In a box there are 5 red and 6 white balls. Aball is drawn out of it and replaced in the box. Then a ball is drawn again. What is the probability that the first ball was red and the second white?

(1) $\frac{1}{30}$ (2) $\frac{30}{121}$ (3) $\frac{1}{121}$ (4) $\frac{5}{121}$ (5) $\frac{1}{11}$

52. From a group of 5 girls and 8 boys, in how many ways can 8 person be chosen so as to include at least one girl?
(1) 984 (2) 1024 (3) 1132 (4) 1242 (5) 1286

53. In a bag there are 9 white and 7 black balls. Two draws of, one ball each are made without replacement. What is the probability that one ball is black and the other is white?
(1) $\frac{19}{40}$ (2) $\frac{21}{40}$ (3) $\frac{19}{56}$ (4) $\frac{37}{56}$ (5) $\frac{24}{95}$

54. If the letters of the word NISHANT are arranged at random, what is the probability that the letters I and A will come together?
(1) $\frac{2}{7}$ (2) $\frac{1}{7}$ (3) $\frac{5}{7}$ (4) $\frac{6}{7}$ (5) None of these

55. How many different numbers of 8 digits can be formed from the digits 0, 1,2,3,4, 5,6,7 that are divisible by 5? (The digits should not be repeated.)
(1) 4320 (2) 5040 (3) 9360 (4) 18720 (5) 4680

56. In how many ways can 6 boys and 5 girls be seated in a row so that they are alternate to one another?
(1) 86400 (2) 518400 (3) 14400 (4) 2880 (5) 30

57. An unbiased coin is tossed 9 times. Find the chances that exactly five tails will appear.
(1) $\frac{21}{256}$ (2) $\frac{63}{256}$ (3) $\frac{15}{256}$ (4) $\frac{63}{128}$ (5) $\frac{21}{128}$

58. A bag contains 9 red, 11 blue and 12 green balls (all the balls are distinct in some manner) . Find out in how many different ways can be three balls selected so that all the three balls are of different colours.
(1) 1138 (2) 1182 (3) 1178 (4) 1188 (5) None of these

59. Seven boys and seven girls are to be seated around a circular table in such a way that no two girls are together. In how many different ways can they be arranged?
(1) 3682880 (2) 3688200 (3) 3628800 (4) 3682200 (5) None of these

60. In how many different ways can we select a queen or a spade from a full pack of cards?
(1) 13 (2) 14 (3) 16 (4) 12 (5) None of these

61. A box contains 3 red, 4 black and 5 green balls. If 3 balls are drawn at random, then what is the probability that they are of the same colour?
(1) $\frac{3}{41}$ (2) $\frac{38}{41}$ (3) $\frac{3}{44}$ (4) $\frac{41}{44}$ (5) None of these

62. In how many different ways can the letters of the word DISPLAY be arranged so that the vowels always come together?
(1) 1440 (2) 720 (3) 5040 (4) 2520 (5) None of these

63. A bag contains 3 red, 4 black and 5 green balls. 13 balls are drawn randomly. What is the probability that the balls drawn are certainly green?
(1) $\frac{5}{41}$ (2) $\frac{5}{44}$ (3) $\frac{7}{41}$ (4) $\frac{7}{44}$ (5) None of these

64. In how many ways can the letters of the word VOLUME be arranged so,, that the vowels always come together?
(1) 720 (2) 360 (3) 180 (4) 144 (5) 24

65. In a party every person shakes hands with every other person. If there were a total of 990 handshakes in the party, find the number of persons who were present in the parry.
(1) 75 (2) 60 (3) 55 (4) 48 (5) 45

66. A and B sit in a round table conference with 15 other persons. If the 17 persons are seated all random, what is the probability that there are exactly 5 persons between A and B?

(1) $\frac{1}{8}$ (2) $\frac{2}{17}$ (3) $\frac{1}{9}$ (4) $\frac{2}{15}$ (5) $\frac{1}{7}$

67. In how many different ways can the letters of the word TRIANGLE be arranged?

(1) 5040 (2) 40320 (3) 43020 (4) 720 (5) 4240

68. A bag contains 6 red and 5 white balls. Three balls are drawn at random. Find the probability that one ball is red and two balls are white.

(1) $\frac{10}{33}$ (2) $\frac{12}{33}$ (3) $\frac{15}{33}$ (4) $\frac{55}{230}$ (5) None of these

Directions (Q. 69-70): Study the given information carefully and answer the questions that follow.

69. A box contains 6 red, 3 yellow and 2 green balls. If two balls are picked at random, what is the probability that both of them are red?

(1) $\frac{3}{11}$ (2) $\frac{1}{2}$ (3) $\frac{4}{15}$ (4) $\frac{1}{7}$ (5) None of these

70. If two balls are picked at random from the box given in the above question, what is the probability that at least one of them is yellow?

(1) $\frac{22}{91}$ (2) $\frac{27}{55}$ (3) $\frac{1}{456}$ (4) $\frac{7}{11}$ (5) None of these

71. A bag contains 7 blue balls and 5 yellow balls. If two balls are selected at random, what is the probability that none is yellow?

(1) $\frac{5}{33}$ (2) $\frac{5}{22}$ (3) $\frac{7}{22}$ (4) $\frac{7}{33}$ (5) $\frac{7}{66}$

72. A die is thrown twice. What is the probability of getting a sum 7 from both the throws?

(1) $\frac{5}{18}$ (2) $\frac{1}{18}$ (3) $\frac{1}{9}$ (4) $\frac{1}{6}$ (5) $\frac{5}{36}$

73. The letters of the word PUMPKIN are arranged in such a way that all the vowels come together. Find out the total number of ways of making such an arrangement.

(1) 360 (2) 5040 (3) 720 (4) 1440 (5) None of these

74. What is the probability that a card drawn at random from a well-shuffled pack of 52 cards is either a king or a diamond?

(1) $\frac{11}{52}$ (2) $\frac{4}{13}$ (3) $\frac{3}{13}$ (4) $\frac{13}{52}$ (5) None of these

75. 5 persons are chosen at random from a group of 4 men, 3 women and 5 children. The probability that exactly 3 of them are children is

(1) $\frac{36}{48}$ (2) $\frac{35}{132}$ (3) $\frac{34}{139}$ (4) $\frac{35}{221}$ (5) $\frac{37}{135}$

76. A box contains 4 white balls, 3 black balls and 9 red balls. In how many ways can 4 balls be drawn from the box, if at least one white ball is to be included in a draw?

(1) 1325 (2) 1421 (3) 325 (4) 428 (5) 912

77. The probability that A can solve a problem is $\frac{2}{3}$, and that B can solve it is $\frac{3}{4}$. If both of them attempt the problem, what is the probability that the problem will be solved?

(1) $\frac{6}{12}$ (2) $\frac{5}{12}$ (3) $\frac{11}{12}$ (4) $\frac{10}{12}$ (5) $\frac{9}{12}$

78. Out of 9 consonants and 5 vowels, how many words of 3 consonants and 2 vowels can be formed?

(1) 100800 (2) 100700 (3) 200800 (4) 200700 (5) 152535

79. From a group of 8 men and 7 women, 6 persons are to be selected to form a committee so that at least 4 men are there on the committee. In how many different ways can this be done?

(1) 2010 (2) 1950 (3) 1720 (4) 1890 (5) 2375

80. If two dice are thrown simultaneously, what is the probability that the sum of the numbers appearing on the top of the dice is less than 5?

(1) $\frac{2}{5}$ (2) $\frac{1}{3}$ (3) $\frac{7}{9}$ (4) $\frac{8}{17}$ (5) None of these

81. In how many ways can 3 boys and 3 girls sit in a row so that neither any two boys nor any two girls sit together?

(1) 720 (2) 36 (3) 72 (4) 12 (5) 35

82. What is the probability that a card drawn from a pack of 52 cards will be either a diamond or a king?

(1) $\frac{2}{13}$ (2) $\frac{4}{13}$ (3) $\frac{1}{13}$ (4) $\frac{1}{52}$ (5) $\frac{2}{37}$

83. Out of 15 students studying in a class, 7 are from Maharastra, 5 from Karnataka and 3 from Goa. 4 students are to be selected at random. What are the chances that at least one is from Karnataka?

(1) $\frac{12}{13}$ (2) $\frac{11}{13}$ (3) $\frac{100}{15}$ (4) $\frac{14}{27}$ (5) $\frac{37}{47}$

84. To fill 12 vacancies there are 25 candidates, of which 5 are from local institutes. 3 vacancies are reserved for the local - institute candidates and the rest are open to all. Find the number of ways in which the selection can be made!

(1) 16796 (2) 167960 (3) 1679600 (4) 176900 (5) 196700

Directions (Q. 85-86): A box contains 9 red, 11 green and 12 yellow balls. Find out in how many ways can 3 balls be selected so that 85.
All 3 balls are of the same colour.

(1) 459 (2) 468 (3) 469 (4) 569 (5) None of these

86. All three balls are of different colours.

(1) 1178 (2) 1188 (3) 1172 (4) 1168 (5) None of these

87. In how many different ways can the letters of the word TRANSPIRATION be arranged so that the vowels always come together?

(1) 2429500 (2) 2721600 (3) 1627800 (4) 2826200 (5) 1360800

88. In a shelf, there are 8 Hindi, 7 English and 6 Urdu books. One book is picked up randomly. What is the probability that it is neither in Hindi nor in Urdu?

(1) $\frac{7}{21}$ (2) $\frac{13}{21}$ (3) $\frac{17}{21}$ (4) $\frac{19}{21}$ (5) $\frac{23}{47}$

89. If the letters of the word MOHAN are arranged at random, what is the probability that all the vowels will come together?

(1) $\frac{2}{5}$ (2) $\frac{3}{5}$ (3) $\frac{4}{5}$ (4) $\frac{3}{4}$ (5) $\frac{1}{2}$

90. In how many different ways can 5 girls and 4 boys be seated in a row so that they are alternate to one another?

(1) 2440 (2) 2660 (3) 2880 (4) 3220 (5) None of these

91. In how many different ways can the letters of the word GENERAL be arranged?

(1) 3280 (2) 2640 (3) 2830 (4) 2520 (5) 2250

92. In how many different ways can the letters of the word COMMENCEMENT be arranged so that the vowels always come together?

(1) 60480 (2) 120960 (3) 122840 (4) 50480 (5) 64080

93. What is the probability of getting a sum 9 from two throws of a dice?

(1) $\frac{1}{8}$ (2) $\frac{1}{7}$ (3) $\frac{2}{5}$ (4) $\frac{1}{9}$ (5) $\frac{2}{7}$

94. A box contains 6 green, 4 blue and 7 red pens. 4 pens are drawn at random. What is the probability that they all are not of the same colour?

(1) $\frac{12}{13}$ (2) $\frac{379}{780}$ (3) $\frac{5341}{6873}$ (4) $\frac{189}{581}$ (5) $\frac{2329}{2380}$

SHORT ANSWER

1. (4)	2. (3)	3. (5)	4. (4)	5. (4)	6. (4)	7. (5)	8. (4)
9. (5)	10. (1)	11. (2)	12. (3)	13. (3)	14. (2)	15. (1)	16. (4)
17. (3)	18. (3)	19. (2)	20. (3)	21. (4)	22. (2)	23. (1)	24. (2)
25. (5)	26. (3)	27. (1)	28. (3)	29. (3)	30. (3)	31. (4)	32. (2)
33. (3)	34. (2)	35. (4)	36. (4)	37. (2)	38. (2)	39. (1)	40. (3)
41. (4)	42. (4)	43. (4)	44. (5)	45. (5)	46. (1)	47. (5)	48. (3)
49. (1)	50. (4)	51. (2)	52. (5)	53. (2)	54. (1)	55. (3)	56. (1)
57. (2)	58. (4)	59. (3)	60. (3)	61. (3)	62. (1)	63. (4)	64. (4)
65. (5)	66. (1)	67. (2)	68. (2)	69. (1)	70. (2)	71. (3)	72. (4)
73. (3)	74. (2)	75. (2)	76. (1)	77. (3)	78. (1)	79. (4)	80. (5)
81. (3)	82. (2)	83. (2)	84. (3)	85. (3)	86. (2)	87. (5)	88. (1)
89. (1)	90. (3)	91. (4)	92. (1)	93. (4)	94. (5)		

DETAIL - EXPLANATIONS

1. 4; As one particular student is included in every selection, we have to choose (5-1) students from (12-1) students. $\therefore$ Total number of selections = ${}^{11}C_4$

$$= \frac{11 \times 10 \times 9 \times 8}{4 \times 3 \times 2} = 330$$

2. 3; There are total eight letters

$\therefore$ n(S) = ${}^{8}P_8$ = 8!

As all vowels should come together, we assume them as one letter. Here E, I, O and U together are taken as one, so the number of letters is 4 + 1 = 5 and it can be arranged in ${}^{5}P_5$ = 5! ways and the vowels can be arranged in 4! ways among themselves $\therefore$ n(E) = 4! × 5!

$\therefore$ P(E) = $\frac{4!5!}{8!}$

$$\therefore P(E) = \frac{4 \times 3 \times 2 \times 5!}{8 \times 7 \times 6 \times 5!}$$

$$= \frac{4 \times 3 \times 2}{8 \times 7 \times 6} = \frac{1}{14}$$

3. 5; Out of the total six letters, three are vowels (E, I, E) and three are consonants (SRS).

Here, E and S come twice. If we take all the vowels together, the number of letters is 3 + 1 = 4

So, letters can be arranged in $\frac{4!}{2!}$ ways and

three vowels can be arranged in $\frac{3!}{2!}$ ways

$\therefore$ Total number of ways $= \frac{4!}{2!} \times \frac{3!}{2!}$

= 12 × 3 = 36

4. 4; From 52 cards, 2 cards are drawn.

$\therefore$ n(s) = ${}^{52}C_2$.

In a pack, there are four queen cards from .which two queen cards have to be drawn.

$\therefore$ n(E) = ${}^{4}C_2$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{{}^{4}C_2}{{}^{52}C_2} = \frac{6}{1326} = \frac{1}{221}$$

5. 4; Total = 28 $\therefore$ Boys = 20, Girls = 8 n(S) = ${}^{28}C_3$ = 3276 n(E) = ${}^{20}C_1 \times {}^{8}C_2$ = 20 × 28 = 560

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{560}{3276} = \frac{140}{819} = \frac{20}{117}$$

6. 4; We assume the six particular persons as one. So, total number of persons is 18 - 6 + 1 = 13 and they can sit around a circular table in 12! ways. Six particular persons can sit in 6! ways among themselves.

$\therefore$ Total number of ways = 12! × 6!

7. 5; Here, the total letters are 7, out of which two are vowels.

Total arrangements without restriction = 7!

If we take the two vowels as one letter, number of letters = 5 + 1 = 6

$\therefore$ Number of arrangements is 6! and the two vowels can be arranged in 2! ways among themselves.

$\therefore$ Total arrangements = 6! × 2!

$\therefore$ Reqd answer = 7! - 6! × 2! = 5040 - (720 × 2) = 5040 - 1440 = 3600 8. 4; Total number of persons is 7 + 6 = 13

$\therefore$ n(S) = ${}^{13}P_{13}$ = 13!

If we take all six girls as one unit, total number of persons = 7 + 1 = 8.

They can be arranged in 8! ways and all the girls can be arranged in ${}^{6}P_6$ = 6! ways among themselves. So, when all the girls are together, total number of arrangements = 8! × 6! Probability of all girls sitting together

$$= P(E) = \frac{8! \times 6!}{13!} = \frac{2}{429}$$

Probability of all girls not sitting together

$$= 1 - \frac{2}{429} = \frac{427}{429}$$

9. 5; Without any restriction, 11 persons can sit in 11! ways. If we take all the girls together as one, the total number of persons will be 7 + 1 = 8, who can sit in 8! ways and the four girls can sit in 4! ways among themselves. So, when all the girls sit together, the number of

ways = 8! × 4! When they don't sit together, the total number of ways = 11! - 8! × 4!

10. 1; Total balls = 4 + 6 + 7 = 17

$\therefore$ n(S) = ${}^{17}C_3$ = 680

Two red balls can be selected from four red balls in ${}^{4}C_2$ = 6 ways, and the third ball can be selected from the remaining 13 balls in ${}^{13}C_1$ = 13 ways.

$\therefore P(E) = \frac{13 \times 6}{680} = \frac{39}{340}$

11. 2; Number of arrangements = 6! × 5!

= 720 × 120 = 86400

12. 3; n(s) = 36 n(E) = 6

$\therefore P(E) = \frac{6}{36} = \frac{1}{6}$

$\because$ Possible combinations are {(4, 1) (1, 4) (5, 2) (2, 5) (6, 3) (3, 6)} = 6

13. 3; There are ten letters in which there are four vowel (O, O, U, I), and 'O' comes twice. If we consider all vowels as one letter the total number of letters will be 6 + 1 = 7 so the number of arrangements = 7!, and four vowels can be arranged in $\frac{4!}{2!}$ ways as 'O' comes twice.

$\therefore$ Total arrangements = $7! \times \frac{4!}{2!}$ = 60480

14. 2;

Man	Woman	Arrangements
5	1	${}^{8}C_5 \times {}^{5}C_1$ = 56 × 5 = 280
4	2	${}^{8}C_4 \times {}^{5}C_2$ = 70 × 10 = 700
3	3	${}^{8}C_3 \times {}^{5}C_3$ = 56 × 10 = 560
2	4	${}^{8}C_2 \times {}^{5}C_4$ = 28 × 5 = 140
1	5	${}^{8}C_1 \times {}^{5}C_5$ = 8 × 1 = 8

$\therefore$ Total = 280 + 700 + 560 + 140 + 8 = 1688

15. 1; One girl is chosen from 3 girls in ${}^{3}C_1$ ways and three boys can be chosen from 7 boys in ${}^{7}C_3$ ways. So, total number of ways

= ${}^{3}C_1 \times {}^{7}C_3$ = 3 × 35 = 105

16. 4; Total cards = 52;

$\therefore$ n(S) = 52 in which there is a total of 13 Diamond cards and four Jack of cards. So, for Diamond n(E) = 13 and for Jack n(E_1) = 4

$\therefore P(E) = \frac{13}{52} = \frac{1}{4}$,

$P(E_1) = \frac{4}{52} = \frac{1}{13}$

As one card is Jack of Diamond, so n($E \cap E_1$) = 1

$\therefore P(E \cap E_1) = \frac{1}{52}$

$\therefore$ Probability of the card being either a Jack or a Diamond.

= $P(E \cup E_1)$

= $P(E) + P(E_1) - P(E \cap E_1)$

$\frac{1}{4} + \frac{1}{13} - \frac{1}{52}$

$= \frac{13 + 4 - 1}{52} = \frac{4}{13}$

17. 3; Let the number of handshakes in the party be ${}^{n}C_2$. $\therefore {}^{n}C_2 = 351$

$\frac{n(n-1)}{2} = 351 \Rightarrow n(n-1) = 702 = 27 \times 26$

$\therefore$ n = 27

18. 3; Total number of balls = 24. Two balls are drawn.

$\therefore$ n(S) = ${}^{24}C_2$ = 276

There are eight black balls, out of which two balls are drawn in ${}^{8}C_2$ ways.

$\therefore$ n(E) = 28 $\therefore P(E) = \frac{28}{276} = \frac{7}{69}$

19. 2; Total number of arrangements

$= \frac{6!}{2!}$ {$\because$ 'O' comes twice}

= 6 × 5 × 4 × 3 = 360

20. 3; There are total four Queen cards and four Jack cards in a pack of 52 cards.

$\therefore P(Queen) = \frac{4}{52} = \frac{1}{13}$, $P(Jack) = \frac{4}{52} = \frac{1}{13}$

$\therefore P(E) = \frac{1}{13} + \frac{1}{13} = \frac{2}{13}$

21. 4; n(S) = ${}^{52}C_2 = \frac{52!}{(52-2)!\,2!} = \frac{52!}{50!\,2!}$

= 26 × 51

There are total four Queens and four Aces in a pack of 52 cards. ∴ $n(E) = {}^4C_1 \times {}^4C_1 = 4 \times 4 = 16$

∴ $P(E) = \frac{16}{26 \times 51} = \frac{8}{663}$

22. 2; There are two possibilities for every students: either to pass or to fail.

∴ Total number of ways = $(2)^9 = 512$

23. 1,

4,5,6		

Ways $\quad {}_4P_2$Ways

∴ Reqd. numbers = $3 \times {}^4P_2 = 3 \times 12 = 36$

24. 2; Sum of the numbers can be 2, 3, 4, 5, and 6.

$n(E) = 15,\ n(S) = 36$ ∴ $P(E) = \frac{15}{36} = \frac{5}{12}$

25. 5; Different possibilities are ∴

Girl	Boy
1	$4 = {}^6C_1 \times {}^8C_4 = 6 \times 70 = 420$
2	$3 = {}^6C_2 \times {}^8C_3 = 15 \times 56 = 840$
3	$2 = {}^6C_3 \times {}^8C_2 = 20 \times 28 = 560$
4	$1 = {}^6C_4 \times {}^8C_1 = 15 \times 8 = 120$
5	$0 = {}^6C_5 \times {}^8C_0 = 6 \times 1 = 6$

∴ Total No. of ways = 1946

26. 3; Probablity that both balls are red

$= \frac{5}{13} \times \frac{5}{13} = \frac{25}{169}$

27. 1; There are nine letters. Taking the three vowels (E, E, A) as a unit, total number of letters becomes 7.

D is repeated twice, and E is repeated twice among vowels.

∴ Total number of arrangements = $\frac{7!}{2!} \times \frac{3!}{2!}$

$= 2520 \times 3 = 7560$

28. 3; n(s) = No. of ways in which seven persons can sit around a round table = 6! As three persons can be taken as a unit, total persons = 5 and number of ways in which five persons can sit around the table = 4! and three persons can sit in 3! ways among themselves.

∴ $n(E) = 4! \times 3!$

∴ $P(E) = \frac{4! \times 3!}{6!} = \frac{1}{5}$

29. 3; Number of handshakes = ${}^nC_2 = 561$

∴ $\frac{n(n-1)}{2} = 561$

∴ $n(n - 1) = 1122$ or $n \times (n - 1) = 34 \times 33$

∴ $n = 34$

30. 3; The sum can be 8, 9, 10, 11 and 12.

∴ $n(E) = 5 + 4 + 3 + 2 + 1 = 15$ and, $n(s) = 36$

∴ $P(E) = \frac{15}{36} = \frac{5}{12}$

31. 4;Total number of balls = 4 + 8 + 12 = 24 n(s) = ${}^{24}C_3 = 2024$

Number of ways to select one ball of each colour

$= n(E) = {}^4C_1 \times {}^8C_1 \times {}^{12}C_1 = 4 \times 8 \times 12 = 384$

∴ $P(E) = \frac{384}{2024} = \frac{48}{253}$

32. 2; Total number of letters is nine, in which N comes twice and total vowels are three (I, U, A).

We consider the three vowels as one letter. So, the total number of letters is 6 + 1 = 7 in which N comes twice. These seven letters can be arranged in $\frac{7!}{2!}$ ways and the three vowels can be arranged in 3! ways.

∴ Total numbers of ways = $\frac{7!}{2!} \times 3! = 15120$

33. 3; (12 - 5) = 7 boys can sit in a row in 7! ways. Now the females can sit in 8 places (including both ends) in ${}^8p_5$ ways.

∴ Total number of ways = ${}^8P_5 \times 7! = \frac{8!}{3!} \times 7!$

$= 6720 \times 5040 = 3386.88 \times 10^4$

34. 2; 12 persons can sit at a round table in (12 1)! = 11! ways. If we take 5 particular persons always together then total number of persons = (12 - 5 + 1) = 8. So, they can sit in 7! ways and those five persons can sit in 5! ways among themselves.

∴ $P(E) = \frac{7! \times 5!}{11!} = \frac{1}{66}$

35. 4; Total number of letters is 7, and these letters can be arranged in 7! ways = 1 × 2 × 3 × 4 × 5 × 6 × 7 = 5040 ways

There are seven letters in the word THERAPY including 2 vowels. (E, A) and five consonants.

Consider two vowels as one letter. We have 6 letters which can be arranged in $^6P_6$ = 6 ways.

But vowels can be arranged in 2! ways. Hence, the number of ways, all vowels will come together = 6! x 2!

= 1 × 2 × 3 × 4 × 5 × 6 × 2 = 1440

Total number of ways in which vowels will never come together = 5040 - 1440 = 3600

36. 4; Total number of balls = 13 + 7 = 20 Number of Sample space = n(S) = $^{20}C_2$ = 190

Number of events = n(E)

= $^{13}C_2$ + $^7C_2$ = 78 + 21 = 99

∴ $P(E) = \frac{n(E)}{n(S)} = \frac{99}{190}$

37. 2;

3,4,5 6or7		

5 Ways $\quad$ $^7P_2$ Ways

∴ = 5 × $\frac{7!}{5!}$ = 5 × 42 = 210

38. 2; n(S) = Total number of ways 15 persons sit at a round table =14!

Taking four persons as one total numbers = 12 persons

∴ Number of ways they can sit at the round table = 11!

Four persons sit among themselves in 4! ways.

So n(E) = 11!4!

∴ $P(E) = \frac{n(E)}{n(S)} = \frac{11!\,4!}{14!} = \frac{1}{91}$

39. 1; When the number is of three digits

5, 6, 7		

3 ways $\quad$ $^5P_2$ ways

∴ Numbers of three digits = 3 × $\frac{5!}{3!}$

= 3 × 5 × 4 = 60

When the number is of four digits

Only digit' 1' can be placed but we don't have that digit So, numbers of 4 digits = 0

40. 3; n(s) = $^{52}C_3$ = 22100

There are total four kings in the pack.

∴ n(E) = $^4C_3$ = 4

∴ $P(E) = \frac{n(E)}{n(S)} = \frac{4}{22100} = \frac{1}{5525}$

41. 4;Total number of letters = 10, vowels = 4, and 'O' comes twice.

We can consider all the vowels as one unit. So, total letters = 7, ie number of arrangements = 7!.

Now vowels can be arranged in 4! ways among them selves and 'O' comes twice.

∴ Total number of ways

= $\frac{7! \times 4!}{2!} = \frac{5040 \times 24}{2}$ = 60480

42. 4; n(s) = $^{13}P_{13}$ = 13!

We can consider all the six boys as one unit. So, total persons will be 7 + 1 and they can be seated in 8! ways and all the six boys can be seated in 6! ways among themselves. So, total number of ways in which all the six boys sit together = $\frac{8! \times 6!}{13!} = \frac{2}{429}$

∴ Probability of all the boys not sitting together

= $1 - \frac{2}{429} = \frac{427}{429}$

43. 4; 1st person will get 2 things from 6 in $^6C_2$ ways, 2nd will get 2 things from the remaining 4 in $^4C_2$ ways and 3rd in $^2C_2$ ways. Total number of ways = $^6C_2$ × $^4C_2$ × $^2C_2$

= 15 × 6 × 1 = 90 ways

44. 5; Total number of balls = 8 + 4 + 3 = 15 n(S) = $^{15}C_3$ = 455 n(E) = $^8C_1$ × $^4C_1$ × $^3C_1$ = 8 × 4 × 3 = 96

∴ $P(E) = \frac{96}{455}$

45. 5; Required number of ways

$^6C_1$ × $^5C_3$ = 6 × 10 = 60 ways

46. 1; Total balls = 6 + 7 + 8 = 21 n(s) = $^{21}C_3$ = 1330

Two blue balls can be selected from 7 blue balls in $^7C_2$ = 21 ways and the remaining one ball can be selected from the remaining 14 balls in $^{14}C_1$ = 14 ways ∴

n(E) = 21 × 14 = 294

∴ $P(E) = \frac{294}{1330} = \frac{147}{665}$

47. 5; The number of letters in the word REPLACE = 7.

The number of arrangements = 7!

But 'E' comes twice.

So, number of arrangements

$\frac{7!}{2!} = 3 \times 4 \times 5 \times 6 \times 7 = 2520$

48. 3; Total number of letters = 7, and 'R' comes twice.

$\therefore$ Number of words = $\frac{7!}{2}$ = 2520

49. 1; Two balls out of 9 can be drawn in $^9C_2$ ways.

$n(S) = \frac{9!}{2!7!} = \frac{9 \times 8}{2} = 36$ ways

One black ball out of 3 balls can be drawn in $^3C_1$ = 3 ways. One white ball out of 6 white balls can be drawn in $^6C_1$ = 6 ways.

n(E) = 3 × 6 = 18 ways

$\therefore P(E) = \frac{18}{36} = \frac{1}{2}$

50. 4; Take the persons of the same nation as one unit. Total three units are there. Total number of arrangements = 3! 6! 5! 3! = 6 × 720 × 120 × 6 = 3110400 51. 2; The first was red and the second white.

$\frac{5 \times 6}{11 \times 11} = \frac{30}{121}$

52. 5; The different possibilities are: girl = 1 + boy = 7

No. ofways = $^5C_1 \times$

$^8C_7$ = 5 × 8 = 40 girl

= 2 + boy = 6

No. ofways = $^5C_2 \times {}^8C_6$

= 10 × 28 = 280 girl =

3 + boy = 5

No. ofways = $^5C_3 \times {}^8C_5$

= 10 × 56 = 560 girl =

4 + boy = 4

No. of ways = $^5C_4 \times$

$^8C_4$ = 5 × 70 = 350 girl

= 5 + boy = 3

No. ofways = $^5C_5 \times {}^8C_3$

= 1 × 56 = 56

$\therefore$ Total number of ways = 40 + 280 + 560 + 350 + 56 = 1286

53. 2; Total balls = 7 + 9 = 16, from which two balls can be drawn in $^{16}C_2$ ways, ie n(s) = 120, One white ball can be drawn from 9 white balls in $^9C_1$ = 9 ways and one black ball can be drawn from 7 black balls in $^7C_1$ = 7 ways. n(E) = 9 × 7 = 63

$\therefore P(E) = \frac{63}{120} = \frac{21}{40}$

54. 1; The total letters are 7, so n(s) = $^7P_7$ = 7! We consider A and I as a unit. So total letters = 5 + 1 = 6. So the number of arrangements = $^6P_6$ = 6! and A and I can be arranged in 2! ways among themselves, so total number of ways = 6! × 2!

$\therefore P(E) = \frac{6! \times 2!}{7!} = \frac{2}{7}$

55. 3; For divisibility by 5 the unit's place should have 0 or 5

(I) When 0 is in unit's place then thearrangement will be in 7! ways, ie 5040 ways.

(II) When 5 is in the unit's place then 0 should not be in the first place. So the toal numbers = 7! - 6! = 5040 - 720 = 4320

Total numbers = 5040 + 4320 = 9360

56. 1;

B	G	B	G	B	G	B	G	B	G

Number of ways = 61 × 5! = 720 × 120 = 86400

57. 2; Reqd probability = $^9C_5 \times \left(\frac{1}{2}\right)^5 \times \left(\frac{1}{2}\right)^4 = \frac{9!}{5!\,4!} \times \frac{1}{32} \times \frac{1}{16} = \frac{63}{256}$

58. 4; All the 3 balls of different colours canbe selected in $^9C_1 \times {}^{11}C_1 \times {}^{12}C_1$ = 9 × 11 × 12 = 99 × 12 = 1188

59. 3; To arrange 7 boys and 7 girls around a circular table so that no two girls sit together, we need to arrange the boys and girls alternately.

$\therefore$ 7 boys in a circular table can be arranged in (7 - 1)! = 6! ways and the 7 girls will be arranged in a linear manner in 7! ways. Therefore, total number of ways = 6! × 7! = 3628800

60. 3; Number of queens = 4

$\therefore$ Number of ways to select one queen = $^4C_1$ = 4 ways

Number of spades =13

$\therefore$ Number of ways to select 1 spade = $^{13}C_1$ = 13 ways.

But, as we know, the set of spades and the set of queens have one card in common the queen of spade. Therefore, it is included twice, and hence needs to be subtracted once while adding both the sets. ∴ Number of ways to select either a queen or a spade

$\therefore n(Q\cup S) = n(Q) + n(S) - n(Q\cap S)$

= 4 + 13 - 1 = 16

Therefore, there are only 16 ways to select a queen or a spade.

61. 3; $n(S) = {}^{12}C_3 = \frac{12!}{9!\,3!}$

$= \frac{12 \times 11 \times 10}{3 \times 2} = 220$

All balls being of same colour can be arranged in ${}^3C_3 + {}^4C_3 + {}^5C_3$ ways.

$\therefore n(E) = {}^3C_3 + {}^4C_3 + {}^5C_3 = 1 + 4 + 10 = 15$

$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{15}{220} = \frac{3}{44}$

62. 1; There are seven letters, of which two are vowels (I, A). We consider these two letters as one {because they should always come together}, so total number of letters will be 7 - 2 + 1 = 6

There is no repetition; hence the total number of arrangements = 6! × 2! = 1440

63. 4;

$n(S) = {}^{12}C_3 = \frac{12!}{3!\,9!} = \frac{12!}{3!\,9!} = \frac{12 \times 11 \times 10}{3 \times 2} = 220$

There is no green ball in the selection, ie 3 balls are selected from 3 red and 4 black balls in ${}^7C_3$ ways.

$\therefore n(E) = \frac{7 \times 6 \times 5}{3 \times 2} = 35$

$\therefore P(E) = \frac{35}{220} = \frac{7}{44}$

64. 4; We consider all three vowels as one.

So, the total number of letters will be 3 + 1 = 4, which can be arranged in 4! ways and three vowels can be arranged in 3! ways among themselves.

Total number of ways of arrangement

1; Total number of balls = 6 + 3 + 2 = 11

Sample space $n(S) = {}^{11}C_2$

Now, the number of events $n(E) = {}^6C_2$

Required probability $P(E) = \frac{n(E)}{n(S)} = \frac{{}^6C_2}{{}^{11}C_2} = \frac{15}{55} = \frac{3}{11}$

69. $n(E) = {}^6C_1 \times {}^6C_1 \times {}^5C_2 = 6 \times 10 = 60$

$\therefore p(E) = \frac{n(E)}{n(S)} = \frac{60}{165} = \frac{12}{33}$

70. 2; At least one is yellow = 1 - P (None is yellow)

$= 1 - \frac{{}^8C_2}{{}^{11}C_2} = 1 - \frac{28}{55} = \frac{27}{55}$

71. 3; Total number of balls = 7 + 5 = 12

Sample space = $n(S) = {}^{12}C_2 = \frac{12!}{10!\,2!} = 66$

Number of events = $n(E) = {}^7C_2$

= 3! × 4! = 6 × 24 = 144

65. 5; The number of handshakes in the party

$\frac{7!}{5! \times 2!} \quad \frac{7 \times 6}{2}$ □ □ □ 21

$\frac{6!}{2!}$

is ${}^{n}C_2$ where n = number of persons.

$\frac{n(n-1)}{2!}$ □ 990

or, n(n - 1) = 1980 or, n^2 - n - 1980 = 0 or, n^2 - 45 + 44n - 1980 = 0 or, n(n - 45) + 44(n - 45) = 0 or, (n - 45) (n + 44) = 0

□ n = 45, -44 (discard - ve value)

Hence, n = 45

72. 4; Sample space n(S) = 6 × 6 = 36

For a sum of 7, events = (1, 6), (6, 1), (5, 2), (2, 5), (4, 3), (3, 4)

n(E) = 6

Probability P(E) = $\frac{n(E)}{n(S)}$ □ $\frac{6}{36}$ □ $\frac{1}{6}$

73. 3; There are two vowels I and U in the given words. They may be treated as a single letter. Thus, there are six letters in which there are two Ps. Hence, the number of arrangements

□ □ □2! 360 2□ □ 720

66. 1; Let the first person A sit on any place. Then there are 16 places for B to sit

So n(s) = 16. If there are exactly 5 persons The two vowels can be arranged in 2! ways between A and B then there can be only 74. 2; Let E and F be the event of getting diamond two possible seats for B, so n(E) = 2 and a king respectively.

P(E) □ $\frac{n(E)}{}$ □ □ $\frac{2}{}$ $\frac{1}{}$ of diamond. n(S) 16 8

Then, E F□ is the event of getting a king

□ n(E) = 13, n(F) = 4 and n(E F□) = 1 67.

2; The number of letters in the word TRIANGLE is 8. Hence (the number of ways) = 8!

So, P(E) = $\frac{13}{52}$ □ $\frac{1}{4}$, P(E)□ $\frac{4}{52}$ □ $\frac{1}{13}$ and

= 1 × 2 × 3 × 4 × 5 × 6 × 7 × 8 = 40320

68. 2; Total number of balls = 6 + 5 = 11

P(E

$\frac{1}{52}$

Let S be the sample space. Then

□ □F)

n(S)□${}^{11}C_3$ □ $\frac{9\ 10\ 11□ □}{1\ 2\ 3□ □}$□165

□ P(a diamond or king)

1 1 1 13 4 1 16 4

4 13 52 52 52 13

75. 2; n(S) = Number of ways of selecting 5

persons out of 12

$= (70 \times 21) + \frac{12}{5!7!} (56 \times 7) + 28 = 1470 + 392 + 28$

= 1890

n(E) = Number of ways of selecting 3 children out of 5, and 2 persons out of (4 + 3 =)

76. n(S) 792 132

77. 1; Reqd number of ways

$= ({}_4C_1 \times {}_{12}C_3) + ({}_4C_2 \times {}_{12}C_2) + ({}_4C_3 \times {}_{12}C_1) + {}_4C_4$

12 11 10 4 3 12 11

4 (4 12) 1

6 2 2

= (4 × 22 × 10) + (6 × 6 × 11) + 48 + 1

= 880 + 396 + 48 + 1 = 1325

3; Problem gets solved = Problem solved . by A but not by B + Problem solved by B but not by A + Problem solved by both = P(A and

B) + P(A × and B) + P(A) P(B) Probability of

7 persons $= {}^5C_3 \times {}^7C_2 = \frac{\lfloor 5}{\lfloor 3 \lfloor 2} \times \frac{\lfloor 7}{\lfloor 5 \lfloor 2} = 210$

$\therefore P(E) = \frac{n(E)}{} = \frac{210}{} = \frac{35}{}$

79. 4; Required number of ways

$= ({}^8C_4 \times {}^7C_2) + ({}^8C_5 \times {}^7C) + ({}^4C_1 \times {}^6C \times {}^7C)_0$

8 7 6 5 7 6 8 7 6 7 8 7

4 3 2 1 2 1 2 3 1 2

$= {}^{12}C_5 = 792$

80. 5; Let the number of favourable events be n(E).

Then n(E) = (1, 1) (1, 2) (1, 3) (2, 2) (2, 1) (3, 1)

$\therefore$ n(E) = 6

$\therefore$ Number of sample space = n(S) = 6 × 6

81. 3; $\therefore$ Reqd probability = P(E) = $\frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}$

Arrangement BGBGBG or GBGBGB

[B → Boys G → Girls]

In first arrangement, boys can sit in 3! ways and girls can sit in 3! ways.

$\therefore$ Total arrangements = 3 × 3 = 6 × 6 = 36 ways

In second arrangement, girls can sit in 3! ways and boys can sit in 3! ways

$\therefore$ Total arrangements = 3 × 3 = 6 × 6

= 36 ways

Hence, required number of ways

Problem solved by A = $P(A) = \frac{2}{3}$

Problem not solved by A = $P(\bar{A}) = 1 - \frac{2}{3} = \frac{1}{3}$

Problem solved by B = $P(B) = \frac{3}{4}$

Problem not solved by B = $P(\bar{B}) = 1 - \frac{3}{4} = \frac{1}{4}$

∴ Problem solved

$= \left(\frac{2}{3}\times\frac{1}{4}\right)+\left(\frac{1}{3}\times\frac{3}{4}\right)+\left(\frac{2}{3}\times\frac{3}{4}\right)$

$= \frac{1}{6}+\frac{1}{4}+\frac{1}{2} = \frac{11}{12}$

78. 1; Number of ways of selecting (3 consonants out of 9) and (2 vowels out of 5)

$= {}^9C_3 \times {}^5C_2 = \frac{9\times8\times7}{3\times2\times1}\times\frac{5\times4}{2\times1} = 840$

Again, each group contains 5 letters. Number of ways of arranging 5 letters among themselves = $\lfloor 5 = 120$

∴ Reqd number of ways = 120 × 840

= 100800

84. 3; Total vacancies = 12 Number of candidates = 25

Number of local candidates = 5

Number of vacancies for local candidates = 3

Which can be filled in ${}^5C_3$ ways.

Remaining vacancies = 12 - 3 = 9

Remaining candidates = 25 - 5 = 20

These vacancies can be filled in ${}^{20}C_9$ ways.

∴ Reqd number of ways = ${}^5C_3 \times {}^{20}C_9$

= 1679600

= (First + Second) = (36 + 36) = 72

82. 2; Here n(s) = 52

There are 13 cards of diamond (including one king) and there are 3 more kings. Let E be the event of getting a diamond, or a king:

Then, n(E) = (13 + 3) = 16

∴ $P(E) = \frac{n(E)}{n(S)} = \frac{16}{52} = \frac{4}{13}$

83. 2; We may have (1 Karnataka student and 3 non-Karnataka students) + (2 Karnataka students and 2 non-Karnataka students) + (3 Karnataka students and I nonKarnataka student) + (4 Karnataka students)

∴ Reqd number of ways = $({}^5C_1 \times {}^{10}C_3) + ({}^5C_2 \times {}^{10}C_2) + ({}^5C_3 \times {}^{10}C_1) + {}^5C_4$

$= \left(5\times\frac{10\times9\times8}{3\times2}\right)+\left(\frac{5\times4}{2}\times\frac{10\times9}{2}\right)+\left(\frac{5\times4}{2}\times10\right)+5 = 1155$

Now, total number of ways = n(S) = ${}^{15}C_4$

= 1365

∴ $P(E) = \frac{n(E)}{n(S)} = \frac{1155}{1365} = \frac{11}{13}$

85. 3; All the three are of the same colour means we choose 3 balls from 9 red or we select 3 balls from 11 green balls Or we select 3 balls from 12 yellow balls. So it can be done in

${}^9C_3 + {}^{11}C_3 + {}^{12}C_3 = 84 + 165 + 220 = 469$ ways

86. 2; All the three balls of different colours can be selected in

${}^9C_1 + {}^{11}C_1 + {}^{12}C_1 = 9 \times 11 \times 12 = 1188$ ways

87. 5; In the word TRANSPIRATION we treat the vowels AIAIO as one letter.

Thus we have TRNSPRTN (AIAIO)

Now we have to arrange 9 letters, out of which T, R, N appear twice and S, P appear once.

∴ Number of ways of arranging these letters

$$= \frac{9!}{2!\,2!\,2!} = 45360$$

Now AIAIO has 5 letters, in which A and I appear twice and O appears once.

∴ Number of ways of arranging these letters

$$= \frac{5!}{2! \times 2!} = 30$$

∴ Reqd number of ways = (45360 × 30) = 1360800

88. 1; Total number of books = 8 + 7 + 6 = 21

Let E be the event that the picked book is neither in Hindi nor in Urdu or the event that the book picked is in English n(E) = 7

$$\therefore P(E) = \frac{7}{21}$$

89. 1; Here, n(S) = 5! n(E) = 4!2!

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{4!\,2!}{5!} = \frac{2}{5}$$

90. 3;

G	B	G	B	G	B	G	B	G

∴ Number of ways = 5! × 4!
= 120 × 24 = 2880

91. 4; The word GENERAL contains 7 letters, and E comes twice.

∴ Reqd number of ways

$$= \frac{7!}{2!} = 7 \times 6 \times 5 \times 4 \times 3 = 2520$$

92. 1; In the word COMMENCEMENT we treat the vowels OEEE as one letter. Thus we have CMMNCMNT (OEEE). We have to arrange 9 letters out of which M occurs 3 times, C occurs 2 times, N occurs 2 times and rest are different letters. Now OEEE has 4 letters in which E occurs 3 times and O only once.

Number of ways of arranging these letters

$$= \frac{9!}{3!\,2!\,2!} \times \frac{4!}{3!} = 60480 \text{ ways}$$

93. 4; In two throws of a dice n(S) = 6 × 6 = 36

Let E = Event of getting a sum 9
= {(3, 6), (4, 5), (5, 4), (6, 3)} n(E)

$$\therefore P(E) = \frac{4}{n(S)} = \frac{4}{36} = \frac{1}{9}$$

94. 5; Let S be the sample space. Then, n(S) = Number of ways of drawing 4 pens out of 17 pens

$$= {}^{17}C_4 = \frac{17 \times 16 \times 15 \times 14}{4 \times 3 \times 2 \times 1} = 2380$$

Let E = Event of drawing 4 pens of the same colour $n(E) = {}^6C_4 + {}^4C_4 + {}^7C_4$

$$= \frac{6 \times 5}{2} + 1 + \frac{7 \times 6 \times 5}{3 \times 2} = 15 + 1 + 35 = 51$$

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{51}{2380}$$

Event of drawing 4 pens such that they all are not of the same colour = 1 - P(E)

$$= 1 - \frac{51}{2380} = \frac{2329}{2380}$$